

Send-It API – Lab 9 Submission

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1. Project Structure

Screenshot 1: VS Code project structure

sendit-api/

├── main.py

├── auth.py

├── seeds.py

├── .env

├── docker-compose.yml

├── alembic.ini

├── uploads/

├── database/

| ├── __init__.py

| └── session.py

├── models/

| ├── __init__.py

| └── user.py

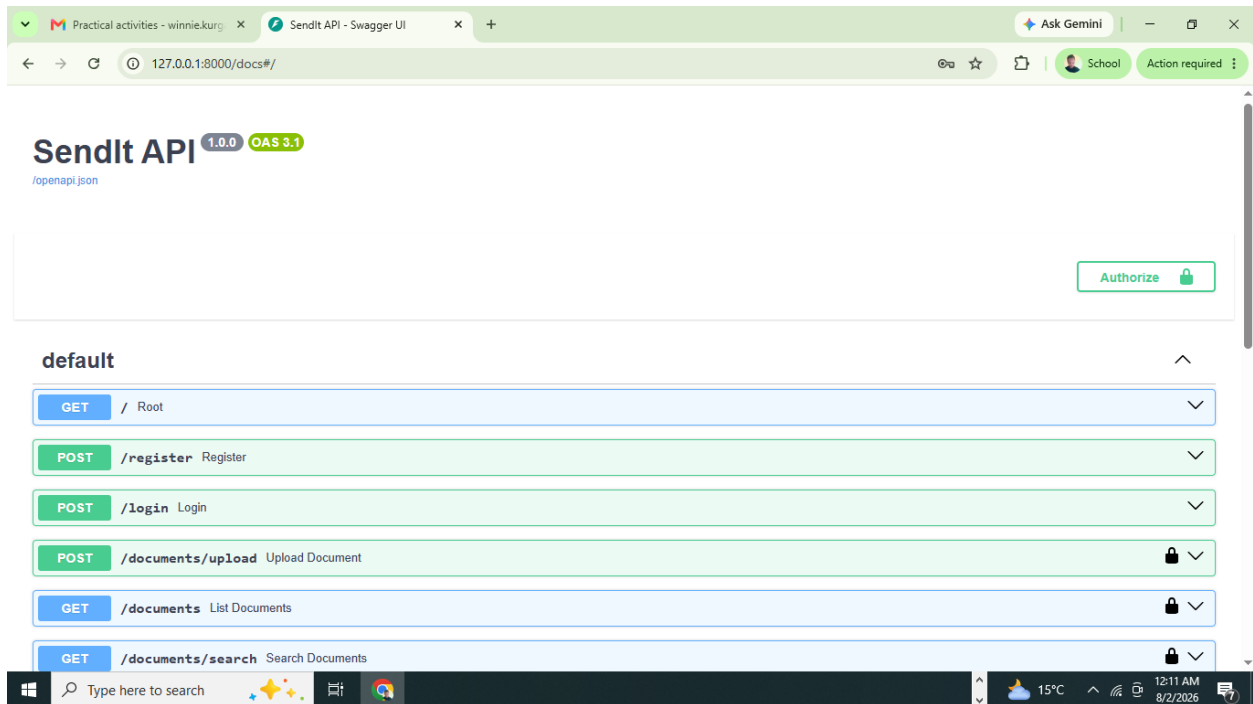
| └── document.py

| └── webhook.py

├── services/

```
| |__init__.py
| |__weather.py
| |__venv/
| |__requirements.txt
```

2. Swagger Documentation



Practical activities - winnie.kurg x SendIt API - Swagger UI x + Ask Gemini | School Action required

127.0.0.1:8000/docs#/

- GET /documents/{document_id} Get Document
- DELETE /documents/{document_id} Delete Document
- POST /documents/{document_id}/enrich Enrich Document
- GET /documents/{document_id}/weather Get Document Weather
- POST /webhooks/register Register Webhook

Schemas

- Body_login_login_post > Expand all object
- Body_upload_document_documents_upload_post > Expand all object
- HTTPValidationError > Expand all object
- UserCreate > Expand all object

Type here to search 15°C 12:11 AM 8/2/2026

3. User Registration

Practical activities - winnie.kurg x SendIt API - Swagger UI x + Ask Gemini | School Action required

127.0.0.1:8000/docs#/default/register_register_post

POST /register Register

Parameters Cancel Reset

No parameters

Request body ^{required} application/json

Edit Value Schema

```
{  "username": "admin",  "email": "admin@example.com",  "password": "Admin1234",  "full_name": "System Admin",  "role": "admin"}
```

Execute Clear

Type here to search Suns... 5:49 PM 8/1/2026

5. Document Upload

The image shows two screenshots of a Swagger UI interface for an API endpoint.

Top Screenshot: POST /documents/upload

Parameters: No parameters.

Request body: multipart/form-data

Form fields:

- file:** required, string. Value: Screenshot (268).png
- city:** required, string. Value: nyeri
- description:** string | (string | null). Value: lab 9 test document
- country:** string. Value: Kenya

Bottom Screenshot: Responses

Curl:

```
curl -X 'POST' \
  'http://127.0.0.1:8000/documents/upload' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJhZG1pbSI6ImV4cCI6MTc4NTU0ODY0MD0.8eBja5SzuGPKIdyGyFocNtVEBuZjZjui0RV7QWixZY' \
  -H 'Content-Type: multipart/form-data' \
  -F 'file=@Screenshot (268).png;type=Image/png' \
  -F 'city=nyeri' \
  -F 'description=lab 9 test document' \
  -F 'country=Kenya'
```

Request URL: http://127.0.0.1:8000/documents/upload

Server response:

Code: 200

Response body:

```
{
  "message": "Document uploaded successfully",
  "document_id": 1,
  "filename": "Screenshot (268).png",
  "status": "enriched"
}
```

Response headers:

```
content-length: 114
content-type: application/json
date: Sat, 01 Aug 2026 15:09:01 GMT
server: uvicorn
```

Figure 5: Successful document upload.

6. List Documents

The screenshot shows a web browser window with the Swagger UI for a SendIt API. The request URL is `http://127.0.0.1:8000/documents`. The server response is a 200 status code with a JSON body. The JSON body contains a single document object with the following fields: `filename`, `original_filename`, `file_type`, `city`, `weather_data`, `description`, `uploaded_at`, `file_path`, `id`, `file_size`, `status`, `country`, `weather_fetched_at`, `uploader_id`, and `updated_at`. The response headers are also displayed, including `content-length`, `content-type`, `date`, and `server`.

Request URL

Server response

Code Details

200 Response body

```
[
  {
    "filename": "20260801_180902_1_Screenshot_(268).png",
    "original_filename": "Screenshot (268).png",
    "file_type": "image/png",
    "city": "nyeri",
    "weather_data": "{\"city\": \"nyeri\", \"country\": \"Kenya\", \"temperature\": 17.1, \"windspeed\": 13.0, \"weathercode\": 51, \"time\": \"2026-08-01T18:00\", \"source\": \"Open-Meteo\"}",
    "description": "lab 9 test document",
    "uploaded_at": "2026-08-01T15:09:02.224474",
    "file_path": "uploads\\20260801_180902_1_Screenshot_(268).png",
    "id": 1,
    "file_size": 85208,
    "status": "enriched",
    "country": "Kenya",
    "weather_fetched_at": "2026-08-01T15:09:05.681782",
    "uploader_id": 1,
    "updated_at": "2026-08-01T15:09:02.225027"
  }
]
```

Response headers

```
content-length: 637
content-type: application/json
date: Sat, 01 Aug 2026 15:10:02 GMT
server: uvicorn
```

Responses

Code	Description	Links
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Figure 6: List of uploaded documents.

7. Search Documents (Exercise 1)

Practical activities - winnie.kurg SendIt API - Swagger UI

127.0.0.1:8000/docs#/default/search_documents_documents_search_get

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/documents/search' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJtYW5hZ2VybWVudCJleHAiOjE3ODU2MjM2MT19.PTj5T02dN76nAZM0bjzHXWFOjCN_Q4BKHHW9aXs'
```

Request URL

http://127.0.0.1:8000/documents/search

Server response

Code Details

200

Response body

```
[
  {
    "version": 1,
    "id": 1,
    "file_type": "image/png",
    "city": "Nairobi",
    "weather_data": "{\"city\": \"Nairobi\", \"country\": \"Kenya\", \"temperature\": 15.8, \"windspeed\": 7.4, \"weathercode\": 3, \"time\": \"2026-08-01T20:27:21.119507\", \"description\": \"lab 9 submission\", \"uploaded_at\": \"2026-08-01T20:27:21.119507\", \"file_path\": \"uploads\\20260801_232721_1_Screenshot_(116).png\", \"original_filename\": \"Screenshot (116).png\", \"filename\": \"20260801_232721_1_Screenshot_(116).png\", \"file_size\": 86267, \"status\": \"enriched\", \"country\": \"Kenya\", \"weather_fetched_at\": \"2026-08-01T20:27:22.942074\", \"uploader_id\": 1, \"updated_at\": \"2026-08-01T20:27:21.120007\"}",
    "description": "lab 9 submission",
    "uploaded_at": "2026-08-01T20:27:21.119507",
    "file_path": "uploads\\20260801_232721_1_Screenshot_(116).png",
    "original_filename": "Screenshot (116).png",
    "filename": "20260801_232721_1_Screenshot_(116).png",
    "file_size": 86267,
    "status": "enriched",
    "country": "Kenya",
    "weather_fetched_at": "2026-08-01T20:27:22.942074",
    "uploader_id": 1,
    "updated_at": "2026-08-01T20:27:21.120007"
  },
  {
    "version": 1,
    "id": 2,
    "file_type": "image/png",
    "city": "nakuru",
    "weather_data": "{\"city\": \"nakuru\", \"country\": \"Kenya\", \"temperature\": 16.5, \"windspeed\": 1.3, \"weathercode\": 3, \"time\": \"2026-08-01T20:27:21.119507\", \"description\": \"lab 9 submission\", \"uploaded_at\": \"2026-08-01T20:27:21.119507\", \"file_path\": \"uploads\\20260801_232721_1_Screenshot_(116).png\", \"original_filename\": \"Screenshot (116).png\", \"filename\": \"20260801_232721_1_Screenshot_(116).png\", \"file_size\": 86267, \"status\": \"enriched\", \"country\": \"Kenya\", \"weather_fetched_at\": \"2026-08-01T20:27:22.942074\", \"uploader_id\": 1, \"updated_at\": \"2026-08-01T20:27:21.120007\"}"
  }
]
```

Figure 7: Document search using filters.

8. Get Document by ID

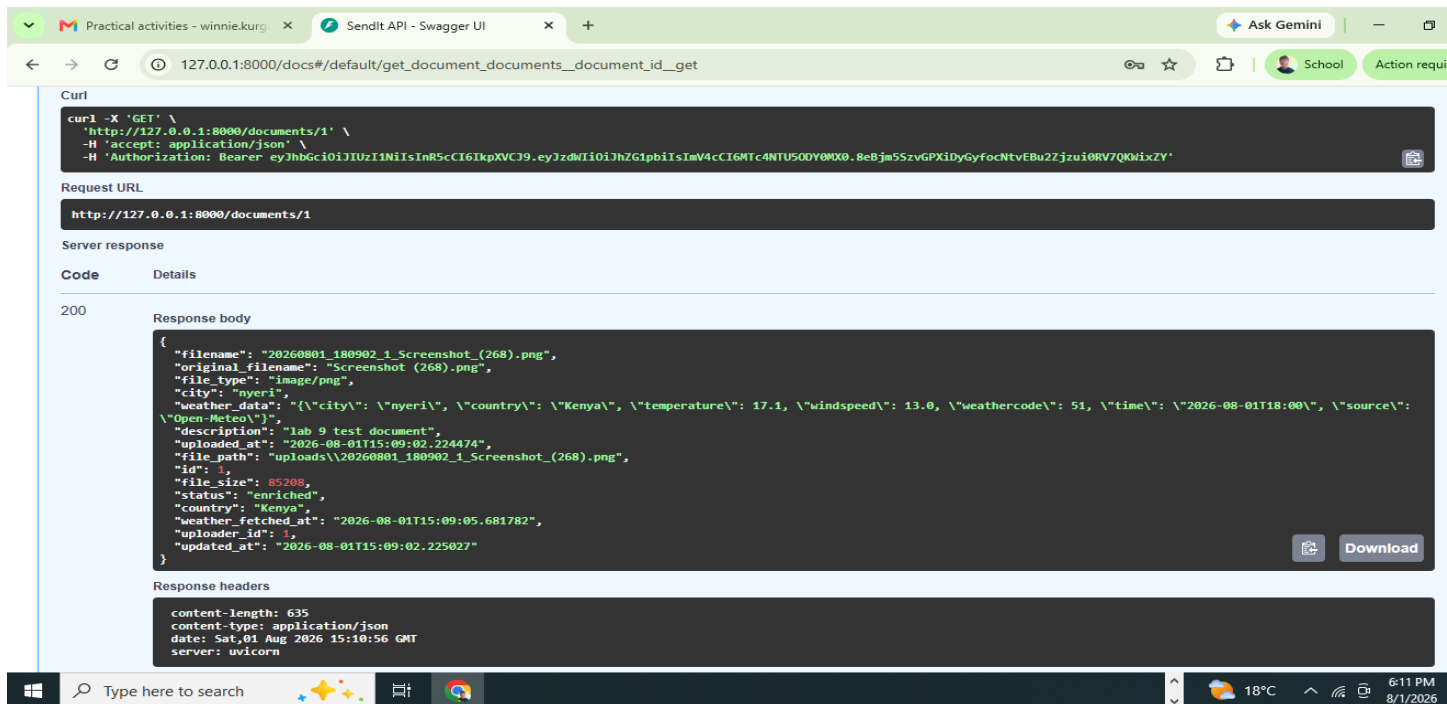
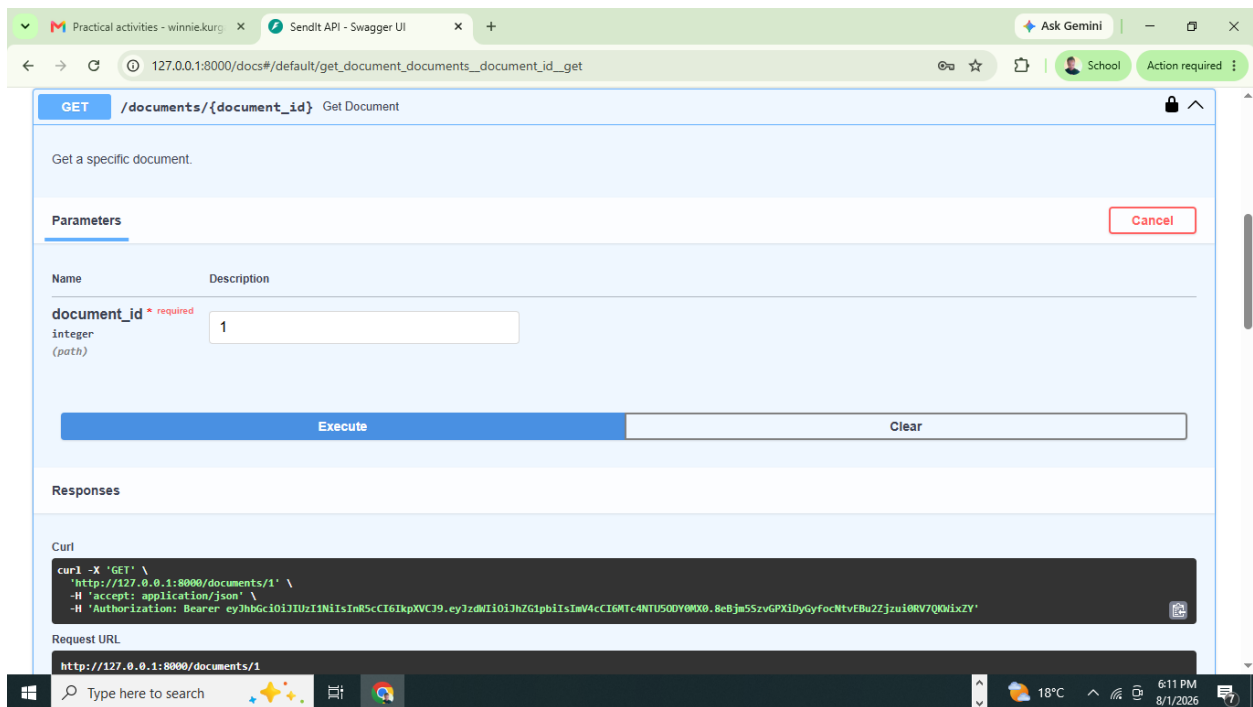


Figure 8: Retrieve document by ID.

9. Weather Enrichment

POST **/documents/{document_id}/enrich** Enrich Document

Manually trigger weather enrichment for a document.

Parameters

Name	Description
document_id * required integer (path)	1

Execute

Responses

Curl

```
curl -X 'POST' \  
  'http://127.0.0.1:8000/documents/1/enrich' \  
  -H 'accept: application/json' \  
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJhZG1pb2IiImV4cCI6MTc4NTU5ODY0MX0.8eBjm5SzvG' \  
  -d ''
```

Request URL

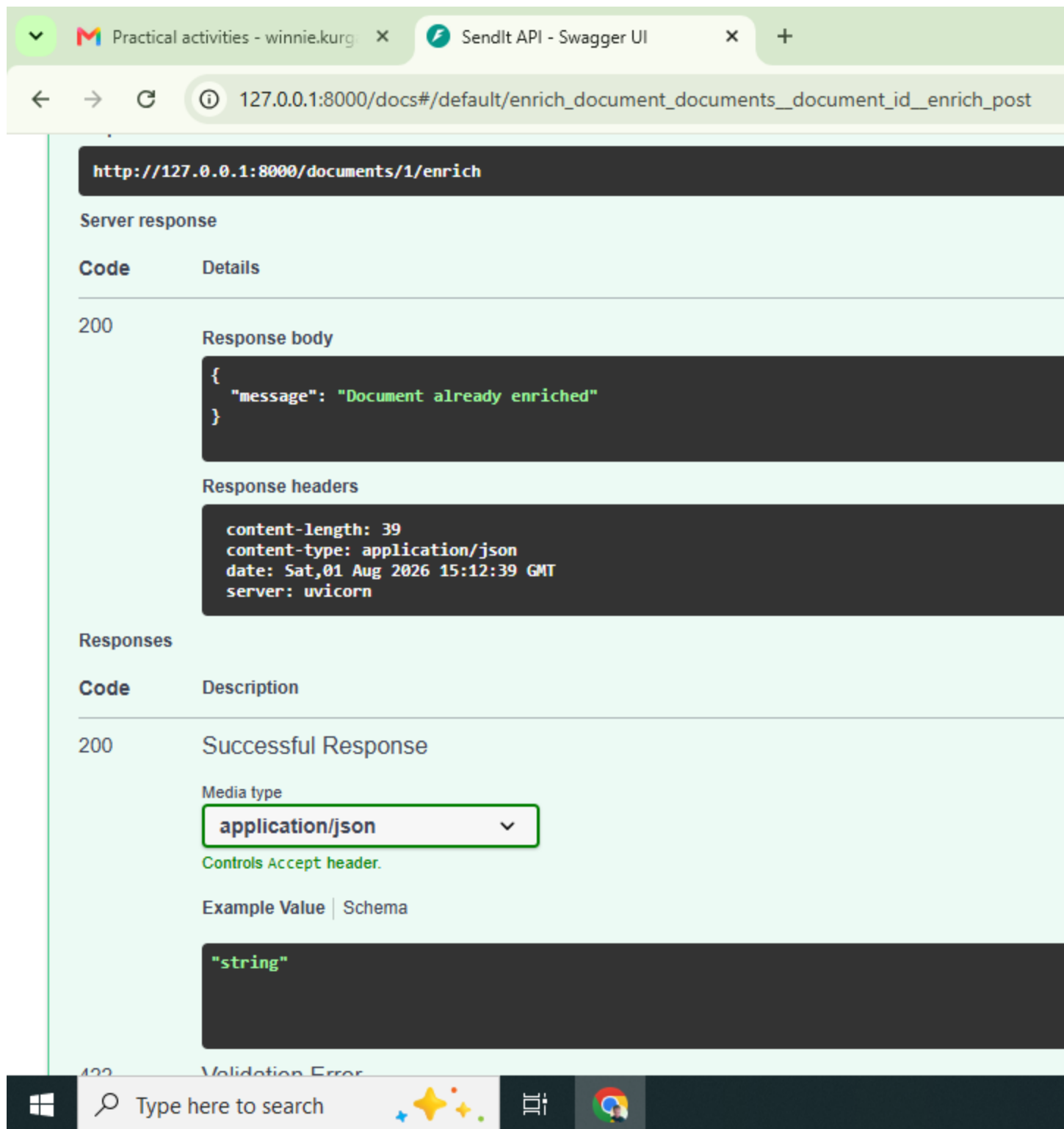


Figure 9: Weather enrichment of a document.

10. Retrieve Weather Information

The image shows two screenshots of a web browser displaying the Swagger UI for an API. The browser tabs include 'Practical activities - winnie.kurg', 'Sendit API - Swagger UI', and 'Ask Gemini'. The address bar shows the URL `127.0.0.1:8000/docs#/default/get_document_weather_documents_document_id_weather_get`.

Top Screenshot: The 'Parameters' tab is active, showing a required integer parameter `document_id` with the value `1`. The 'Execute' button is highlighted in blue. Below the parameters, the 'Responses' section shows a cURL command and the request URL `http://127.0.0.1:8000/documents/1/weather`.

Bottom Screenshot: The 'Responses' section is expanded, showing a successful `200` response. The response body is a JSON object:

```
{
  "document_id": 1,
  "city": "Nyeri",
  "country": "Kenya",
  "weather": {
    "city": "Nyeri",
    "country": "Kenya",
    "temperature": 17.1,
    "windspeed": 12,
    "weathercode": 51,
    "time": "2026-08-01T18:00",
    "source": "Open-Meteo"
  }
}
```

The response headers are also displayed:

```
content-length: 196
content-type: application/json
date: Sat, 01 Aug 2026 15:12:02 GMT
server: unicorn
```

At the bottom, a table lists the response details:

Code	Description	Links
200	Successful Response	No links

The media type is set to `application/json`.

Figure 10: Weather information for a document.

11. Delete Document

The screenshot displays the Swagger UI interface for a REST API. The browser's address bar shows the URL `127.0.0.1:8000/docs#/default/delete_document_documents__document_id_delete`. The API endpoint is `DELETE /documents/{document_id}` with the description "Delete a document (Managers and Admins only)".

Parameters

Name	Description
document_id * required integer (path)	1

Buttons: **Execute** (blue) and **Clear** (white).

Responses

Curl

```
curl -X 'DELETE' \
'http://127.0.0.1:8000/documents/1' \
-H 'accept: application/json' \
-H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJhZG1pbSI6ImV4cCI6MTc4NTU5ODY0eX0.8eBjw55zvGPXI0yGyfoCttvEBu2Zjzui0RV7Q0ixZY'
```

Request URL

```
http://127.0.0.1:8000/documents/1
```

The bottom of the image shows a Windows taskbar with the search bar, task view button, and system tray displaying the temperature (18°C), time (6:13 PM), and date (8/1/2026).

Practical activities - winnie.kurg x SendIt API - Swagger UI +

127.0.0.1:8000/docs#/default/delete_document_documents__document_id__delete

Server response

Code	Details
200	Response body <pre>{ "message": "Document deleted successfully" }</pre> Response headers <pre>content-length: 43 content-type: application/json date: Sat, 01 Aug 2026 15:13:15 GMT server: uvicorn</pre>

Responses

Code	Description
200	Successful Response Media type application/json v Controls Accept header. Example Value Schema <pre>"string"</pre>
422	Validation Error Media type

Windows taskbar: Type here to search, task view, Google Chrome

Figure 11: Successful document deletion.

12. Webhook Registration (Exercise 3)

Practical activities - winnie.kurg

SendIt API - Swagger UI

127.0.0.1:8000/docs#/default/register_webhook_webhooks_register_post

POST

/webhooks/register

Register Webhook

Parameters

Name	Description
webhook_url * required string <i>(query)</i>	https://example.com/webhook
event_type * required string <i>(query)</i>	document.enriched

Execute

Responses

Curl

```
curl -X 'POST' \
'http://127.0.0.1:8000/webhooks/register?webhook_url=https%3A%2F%2Fexample.com%2Fwebhook&event_type=document.enriched' \
-H 'accept: application/json' \
-H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJhZG1pbSI6ImV4cCI6MTc4NTYxMjEwMDAwfQ.eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJhZG1pbSI6ImV4cCI6MTc4NTYxMjEwMDAwfQ' \
-d ''
```

Request URL

Practical activities - winnie.kurg x SendIt API - Swagger UI x +

127.0.0.1:8000/docs#/default/register_webhook_webhooks_register_post

Responses

Curl

```
curl -X 'POST' \
'http://127.0.0.1:8000/webhooks/register?webhook_url=https%3A%2F%2Fexample.com%2Fwebhook&event_type=' \
-H 'accept: application/json' \
-H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJhZG1pb2IiImV4cCI6MTc4NTYx' \
-d ''
```

Request URL

```
http://127.0.0.1:8000/webhooks/register?webhook_url=https%3A%2F%2Fexample.com%2Fwebhook&event_type=
```

Server response

Code	Details
200	Response body <pre>{ "message": "Webhook registered successfully", "webhook_id": 1, "event_type": "document.enriched" }</pre> Response headers <pre>content-length: 93 content-type: application/json date: Sat, 01 Aug 2026 19:58:21 GMT server: uvicorn</pre>

Responses

Code	Description
200	Successful Response

Windows taskbar: Type here to search, task view, Google Chrome

Figure 12: Successful webhook registration.

13. Exercise 2 – Document Versioning

POST /documents/upload Upload Document

Upload a document with validation. Enriches the document with weather data.

Parameters

No parameters

Request body required

file * required

string

Choose File Screenshot (268).png

city * required

string

nyeri

description

string |
(string | null)

lab 9 submission

☐ Send empty value

country

string

Kenya

☐ Send empty value



Type here to search



Practical activities - winnie.kurg SendIt API - Swagger UI

127.0.0.1:8000/docs#/default/upload_document_documents_upload_post

http://127.0.0.1:8000/documents/upload

Server response

Code	Details
200	Response body <pre>{ "message": "Document uploaded successfully", "document_id": 6, "filename": "Screenshot (268).png", "version": 2, "status": "enriched"}</pre> Response headers <pre>content-length: 126 content-type: application/json date: Sun, 02 Aug 2026 08:07:22 GMT server: uvicorn</pre>

Responses

Code	Description
200	Successful Response Media type application/json Controls Accept header. Example Value Schema <pre>"string"</pre>

Windows taskbar with search bar and application icons.

Figure 13: Document version increment after uploading the same file twice.

14. GitHub Repository

<https://github.com/winnie-Jemutai/sendit-api>

Figure 14: SendIt API GitHub repository.

Exercise Answers

Exercise 1 – Search Functionality

1. How would you make document search efficient?

- Create database indexes on commonly searched fields.
- Use pagination to limit results.
- Optimize SQL queries.

2. Should managers see all documents? Explain.

Yes. Managers and administrators should have permission to view all documents for monitoring, auditing, and management purposes, while regular staff should only access their own documents.

Exercise 2 – Document Versioning

1. How would you track document versions?

Store a version number for each uploaded document and increment it whenever a document with the same filename is uploaded again.

2. Should previous versions be deleted? Explain.

No. Previous versions should be retained to provide an audit trail, enable rollback, and preserve document history.

Exercise 3 – Webhooks

1. How would you handle failed webhook deliveries?

Retry failed deliveries using exponential backoff, log failures, and stop after a defined maximum number of retries.

2. What security measures would you implement for webhooks?

- HTTPS encryption

- HMAC request signatures
- Authentication/authorization
- URL validation
- Rate limiting